Literature review of time series anomaly detection methods

Anomaly detection in time series data is a critical research area with numerous applications, including Cybersecurity, Predictive maintenance, Fraud detection, Industrial damage detection, Data leak prevention, Healthcare , Finance, Industrial monitoring [8], etc.

Anomaly detection in this context involves identifying data points or sequences that deviate significantly from expected patterns. Conventional anomaly detection techniques, like Statistical methods, Regression models, rule-based techniques etc., rely on static and time-invariant models and cannot efficiently tackle the anomaly detection problem. The main limitations of conventional methods are:

- Difficulty Handling High-Dimensional Data: Many conventional methods struggle to effectively analyze high-dimensional time series data. This is particularly true for distance-based methods, which can be affected by the "curse of dimensionality," where the distance between points becomes less meaningful as the number of dimensions increases [1-3]. Dimensionality reduction techniques are often necessary to address this issue, but they can lead to information loss [4-6].
- Inability to Capture Complex Temporal Dependencies: Some conventional techniques, especially those based on statistical methods, often assume linearity and stationarity in time series data. They may not be able to accurately model complex temporal dependencies, leading to inaccurate anomaly detection [7-11].
- Limited Ability to Detect Complex Anomalies: Many conventional methods are primarily designed to detect point anomalies, where a single data point deviates sharply from expected behavior. They may not be as effective in identifying contextual anomalies, which depend on the specific context or time of occurrence, or collective anomalies, which involve a sequence of data points exhibiting abnormal behavior [12-18].
- Dependence on Parameter Tuning: Several conventional methods rely on setting appropriate parameters or thresholds to define normal behavior and identify anomalies. This can be challenging and often requires domain expertise. Improper parameter settings can lead to poor detection performance [3, 19-22].
- Lack of Explainability: Many conventional techniques are considered "black boxes," meaning they provide little insight into why a particular data point is flagged as an anomaly. This lack of explainability can hinder the ability to understand the root causes of anomalies and take appropriate actions [23, 24].

These limitations have driven the research community to explore and develop more advanced anomaly detection methods, particularly those based on deep learning. These newer techniques leverage the ability of neural networks to learn complex patterns and relationships, enabling them to potentially overcome some of the limitations of conventional approaches. Many researchers have conducted detailed reviews of advanced anomaly detection methods for readers to refer to [7 – 19]. Existing methods can be categorized based on their underlying approach:

- Statistical and probabilistic methods [5] utilize historical data to establish a model of expected system behavior. Anomalies are flagged when new observations deviate from this model. Examples include AR, MA, ARMA, and ARIMA models [9].

- Classical machine learning approaches [10], such as clustering (e.g., k-means) and support vector machines (SVMs) [11], aim to detect anomalies without assuming a specific data generation process.
- Deep learning techniques [12] leverage the power of neural networks to detect anomalies. Architectures like Long Short-Term Memory (LSTM) networks, autoencoders, and Generative Adversarial Networks (GANs) are often employed. [7, 13, 14]

Several studies focus on evaluating and comparing the performance of these methods, often using benchmark datasets [15-17]. Factors like accuracy, computational efficiency, and robustness are considered in these evaluations [15, 18]. Other studies identify various challenges inherent to anomaly detection in time series data, including:

- The need for real-time frameworks to handle data streams and address the demands of real-world systems [19].
- Dimensionality reduction is often required to manage high-dimensional time series data efficiently [1].
- Anomaly localization involves pinpointing the precise location of anomalies within the time series [1].
- Explainability of anomalies remains a significant hurdle. While many studies focus solely on detection methods, understanding the underlying reasons for anomalies is crucial for effective mitigation [14, 20].

Studies also highlight the significance of selecting appropriate evaluation metrics for anomaly detection, especially for range-based anomalies that cover multiple observations [21]. Traditional point-based measures, such as precision, recall, and F1-score, may not be suitable in these situations.

Furthermore, researchers underscore that the choice of anomaly detection technique depends on factors like data type, anomaly type, application domain, and specific requirements [22-25]. They also highlight the need for future research to address the identified challenges and advance the field of time series anomaly detection [20, 26-29].

Most popular time series anomaly detection methods

The detailed reviews [7 – 19] of existing anomaly detection methods have revealed some of the most popular and efficient approaches, which are described here.

LSTM-Based Methods for Anomaly Detection

LSTM networks are well-suited for anomaly detection in time series data due to their ability to learn temporal dependencies and capture them in a low-dimensional state representation [1, 2]. Some of their advantages are the following:

- They can effectively detect contextual anomalies, which are data points considered abnormal based on their surrounding context [2-4].
- They can handle both stationary and non-stationary dynamics and short-term and long-term dependencies [2].

LSTMs can be combined with other techniques, such as encoder-decoder architectures and hybrid approaches, to improve detection capabilities further [1, 5]. For example, encoder-decoder-based LSTMs can optimize detection abilities in high-dimensional data spaces by learning a compressed representation of

normal data and identifying anomalies based on reconstruction errors [5, 6]. Hybrid approaches can combine the benefits of LSTMs with other methods, such as incorporating a predictor and detector component to divide tasks effectively [5].

However, LSTM-based methods also have limitations:

- They can be computationally intensive and time-consuming to train, especially with long time series data [1, 7].
- Despite their ability to mitigate the vanishing gradient problem, it can still occur, particularly with very long sequences, hindering the learning of long-term dependencies.
- Their performance can be sensitive to hyperparameter settings, making tuning for optimal performance a challenge.

Autoencoder-Based Methods for Anomaly Detection

Autoencoders (AEs) are another popular approach for anomaly detection. They operate by learning a compressed representation of normal data and then reconstructing the input. Anomalies are detected based on the reconstruction error, which is expected to be higher for abnormal data points [8, 9]. Various types of autoencoders exist, including:

- Denoising Autoencoders: These are designed to be robust to noise by corrupting the input data during training and learning to reconstruct the original, clean data [3, 10].
- Robust Deep Autoencoders: These combine autoencoders with robust principal component analysis (RPCA) to isolate noise and outliers, allowing for accurate reconstruction of normal data [10].
- Variational Autoencoders (VAEs): These introduce stochasticity by learning a probabilistic distribution of the latent space, enabling them to model unseen patterns and generate new data similar to the training data [8, 11].
- Contractive Autoencoders: These encourage the network to learn a representation that is robust to small perturbations in the input data [5].
- Variational Quasi-Recurrent Autoencoders (VQRAEs): These combine the benefits of VAEs with quasi-recurrent neural networks to achieve both temporal modeling and computational efficiency [12, 13].

Despite their effectiveness, AE-based methods also face challenges:

- Determining the optimal threshold for the reconstruction error to classify anomalies can be challenging.
- AEs can struggle with high-dimensional data and are sensitive to noise, requiring techniques like feature selection, dimensionality reduction, or denoising to improve performance.

GAN-Based Methods for Anomaly Detection

Generative Adversarial Networks (GANs) are a relatively new class of deep learning models that have shown promising results in anomaly detection. GANs consist of two components: a generator that learns to generate synthetic data similar to real data and a discriminator that learns to distinguish between real and synthetic data [3, 14]. For anomaly detection, GANs are trained on normal data, and anomalies are detected based on

the discriminator's output or the reconstruction error of the generator [14, 15]. There are several benefits of using GANs:

- They can capture complex data distributions and generate synthetic examples of normal data, making them suitable for handling high-dimensional and complex datasets.
- They can be used in unsupervised or semi-supervised settings where labeled anomalies are scarce.

However, GAN-based methods also have some drawbacks:

- Training GANs can be notoriously unstable and challenging, often encountering issues like mode collapse or non-convergence.
- The interpretability of GANs is limited, making it difficult to understand why a particular data point is flagged as anomalous.

GNN-Based Methods for Anomaly Detection

Graph Neural Networks (GNNs) are particularly well-suited for anomaly detection in graph-structured data. GNNs learn node representations by aggregating information from neighboring nodes, enabling them to capture the structural properties and relationships within the graph [16, 17]. The most noticeable advantages of GNNs for anomaly detection are the following:

- They can effectively detect various types of graph anomalies, including structural, global, and community anomalies.
- They can leverage both node attributes and graph structure for improved detection accuracy.
- GNNs are adaptable to different graph types, such as attributed graphs, dynamic graphs, and multi-relational graphs.

Despite their strengths, GNN-based methods also face limitations:

- They can struggle with scalability when dealing with large graphs containing millions of nodes and edges, requiring efficient algorithms and data structures.
- Acquiring sufficient labeled data for training GNNs can be difficult in real-world scenarios.
- Current GNN-based methods predominantly focus on detecting specific types of anomalies, with limited research on collective and graph-level anomalies.

Other Methods for Anomaly Detection

Other methods widely used for anomaly detection on time-series are:

- RNN-Based Methods: These are similar to LSTMs but are generally less effective in learning long-term dependencies and can also suffer from the vanishing gradient problem.
- Contrastive Learning Methods: These learn representations by distinguishing similar views of the same sample from other data samples. However, they are sensitive to negative sampling strategies and often rely on two-stage approaches, which may not align well with the anomaly detection task [18-21].

While the aforementioned deep learning methods like LSTMs, AEs, GANs, and GNNs offer promising results in handling complex data and capturing intricate relationships, they also present challenges regarding computational cost, training instability, and interpretability. Researchers continue to explore new approaches and refine existing methods to address these limitations and develop more accurate, efficient, and explainable anomaly detection systems.

A novel approach. The Kolmogorov-Arnold Networks (KANs)

A multifaceted examination of KANs, exploring their theoretical foundations, structural advantages, various implementations, and potential applications can be found in […].

At the heart of KANs lies the Kolmogorov-Arnold representation theorem ([1-6]). This theorem fundamentally asserts that any continuous multivariate function can be expressed as a composition of univariate functions. KANs leverage this principle by replacing the traditional linear weight multiplications in MLPs with learnable univariate functions positioned on the connections between nodes. These functions effectively act as activation functions, dynamically adapting during the training process.

KANs present a significant departure from conventional MLP architectures, offering several potential advantages ([1, 3, 4, 6]):

- Enhanced Interpretability: Unlike the often-opaque nature of MLPs, the univariate functions in KANs provide a clearer understanding of the model's internal workings ([1, 3, 6, 7]). This transparency makes it easier to interpret the model's decision-making process and potentially uncover insights into the underlying data patterns.
- Improved Accuracy: KANs exhibit the capacity to achieve higher accuracy with fewer parameters compared to MLPs ([1, 3, 6, 8]). This efficiency stems from their ability to effectively approximate complex functions using adaptable univariate functions, typically parameterized using B-splines ([1, 3]) or, in some variants, wavelet functions ([9]).
- Increased Efficiency: The reduced number of parameters in KANs translates to faster training times and lower computational demands ([1, 3, 6]). This efficiency makes KANs potentially well-suited for applications where computational resources are limited or real-time processing is required.

There are several specialized KAN architectures proposed by researchers, each tailored to address specific challenges or leverage particular data characteristics ([8-12]):

- TKAN (Temporal KAN): Introduced by Genet and Inzirillo ([10, 13]), this variant incorporates recurring and gating mechanisms akin to LSTMs to effectively capture long-term dependencies within time series data. TKAN's design aims to improve performance in multi-step prediction tasks, as demonstrated in experiments with cryptocurrency market data ([10, 13]).
- Wav-KAN: Bozorgasl and Chen ([9, 14]) propose this architecture that utilizes wavelet functions instead of the more common B-spline parameterization. They argue that wavelets, known for their ability to efficiently capture both high-frequency and low-frequency components in data, can lead to faster training and enhanced accuracy, particularly in scenarios with irregular sampling or noise.

- T-KAN and MT-KAN: Specifically designed for time series forecasting, these variants are presented by Xu, Chen, and Wang ([12, 15]). T-KAN focuses on univariate time series and incorporates symbolic regression to further enhance interpretability by deriving explicit mathematical relationships from the learned activation functions. MT-KAN extends this approach to multivariate time series, aiming to capture complex interactions between multiple variables for improved predictive performance.

Current research highlights the potential of KANs across a diverse range of applications, including:

- Time Series Forecasting: KANs have demonstrated promising results in forecasting tasks, particularly in financial applications ([12, 15, 16]). Their ability to model complex nonlinear relationships and adapt to changing patterns makes them well-suited for predicting volatile time series like stock market data.
- Satellite Traffic Forecasting: Vaca-Rubio et al. ([8, 17]) showcase the effectiveness of KANs in predicting satellite traffic patterns. Their experiments demonstrate that KANs can outperform traditional MLPs, achieving higher accuracy with fewer parameters. This efficiency is crucial in satellite communication networks where resources are often constrained.
- Computational Biomedicine: Samadi et al. ([11, 18]) suggest that "smooth, structurally informed KANs" could be particularly beneficial in computational biomedicine. They argue that by incorporating prior knowledge about the biological systems under study, KANs could reduce the amount of training data required, enhance model reliability, and facilitate extrapolation to sparsely sampled data regions.

Despite their promise, the current state-of-the-art acknowledge that KANs are a relatively new development, and further research is needed to address existing challenges and fully realize their potential ([1, 3, 11, 19-21]). Some indicative fields of further research are the following:

- Smoothness Considerations: The choice of univariate functions and their smoothness properties can significantly impact the representational capacity of KANs ([11, 22]). Achieving a balance between smoothness, which can aid in generalization, and the ability to capture complex data patterns requires careful consideration and further investigation.
- Network Topology Optimization: Determining optimal network topologies for different tasks remains an open challenge in KAN research ([1, 19]). Developing methods for automatically adapting the topology based on data characteristics or incorporating domain knowledge could significantly improve performance.
- Scalability to Large Datasets: While KANs have shown promise in smaller-scale experiments, their scalability to larger datasets and more complex tasks needs further exploration ([1, 19]). Efficient training algorithms and techniques for handling high-dimensional data are crucial for broader adoption of KANs.
- Integration with Existing Architectures: Exploring the integration of KANs with other established architectures like RNNs, LSTMs, and Transformers could lead to synergistic improvements ([1, 23]). Combining the interpretability and efficiency of KANs with the strengths of these architectures could create powerful hybrid models for various applications.
- Addressing the Black-Box Nature of Neural Networks: The need for interpretability in AI systems is a driving force behind the development of KANs ([20]). Understanding the decision-making process of neural networks is important to building trust and ensuring responsible deployment, particularly in critical domains like healthcare and finance.

Potential Approaches for Time Series Anomaly Detection Using KANs

While the current research mainly focuses on the application of KANs for time series forecasting, they do not explicitly address their use for anomaly detection. However, based on the characteristics of KANs, potential approaches for applying them to anomaly detection tasks can be explored.

Some potential approaches for anomaly detection are:

- Reconstruction-Based Anomaly Detection: Train a KAN to model the normal behavior of a time series.
- Use the trained KAN to reconstruct the input time series.
- Calculate the reconstruction error (e.g., using MSE or MAE) between the original and reconstructed series.
- Large reconstruction errors could indicate anomalies, as the KAN would struggle to model deviations from the learned normal patterns accurately.
- Prediction-Based Anomaly Detection: Train a KAN to predict future time series values.
- Compare the predicted values with the actual observations.
- Significant deviations between predictions and actual values could indicate anomalies.
- Interpretable Feature Extraction for Anomaly Detection: Utilize the interpretable nature of KANs to extract meaningful features from the time series.
- The learned activation functions could capture distinct patterns and trends in the data.
- Anomalies could be identified by analyzing these features, potentially using traditional anomaly detection techniques or establishing thresholds for specific feature values.

Review of feature extraction from time series

There is a consensus among researchers that made reviews of anomaly detection methods [7 – 19] about the importance of feature engineering in time series anomaly detection. This process involves extracting relevant features from the raw time series data to improve the performance of anomaly detection algorithms. Feature extraction can also be used to reduce the dimensionality of high-dimensional time series data. Techniques like PCA can be applied after extracting representative statistical features (e.g., mean, autocorrelation) from each time series. Selecting the right features for anomaly detection is crucial but challenging. Irrelevant features can hinder the performance of anomaly detection algorithms.

Generally, time series features can be grouped into:

- Time-domain features: These features are derived directly from the time series values and include statistical measures (e.g., mean, variance, skewness), temporal measures (e.g., autocorrelation, partial autocorrelation), and measures of variability (e.g., moving average, exponential smoothing).
- Frequency-domain features: These features are obtained by transforming the time series data into the frequency domain using techniques like the Fourier Transform. They capture periodic patterns and trends in the data.
- Wavelet-based features: Wavelet analysis provides a time-frequency representation of the data and can be used to extract features that capture both temporal and frequency characteristics.

Highly comparative time-series analysis (HCTSA)

hctsa (Highly Comparative Time-Series Analysis) is a software tool designed for automated time-series phenotyping using massive feature extraction. The software was developed by Ben D. Fulcher and Nick S. Jones. It is based on their previous work on highly comparative, feature-based time series classification.

The main functionality of hctsa can be summarized in the following:

- Massive Feature Extraction: The core of hctsa lies in its ability to extract over 7,700 features from time series data. These features come from a wide range of scientific disciplines, including basic statistics, correlation structures, entropy measures, stationarity estimates, model fits, and methods from nonlinear time-series analysis.
- Automated Feature Selection: hctsa includes algorithms for automatically selecting the most informative features for a given application. This circumvents the need for manual selection, which can be time-consuming and subjective.
- Classification and Regression: hctsa can be used to build classifiers and regression models for time series data. It offers tools for both supervised and unsupervised learning.
- Visualization: hctsa provides various visualization tools to help users understand the structure of their data and the relationships between different features.

Hctsa has been used in a wide range of applications:

- Phenotyping: hctsa is particularly well-suited for phenotyping applications, where the goal is to identify meaningful quantitative phenotypes from time-series data. The authors demonstrate this using movement speed time series of C. elegans and Drosophila.
- Disease Diagnosis: hctsa can be used to diagnose diseases from biomedical time series such as heart rates or brain dynamics. This is demonstrated in a case study distinguishing between "normal sinus rhythm" and "congestive heart failure" heartbeat interval series.
- Other Applications: hctsa has broader applications in various fields that involve time series analysis, including but not limited to cognitive science, industrial process monitoring, and financial market analysis.

The main advantages of hctsa can be summarized in the following:

- Leveraging Existing Knowledge: hctsa allows researchers to leverage decades of research in time-series analysis by providing access to a vast library of features.
- Data-Driven Analysis: hctsa promotes a data-driven approach to time-series analysis by automating feature selection and model building.
- Interpretability: While hctsa employs advanced techniques, it emphasizes interpretability by selecting features that have clear scientific meanings.
- Dimensionality Reduction: By representing time series using a reduced set of features, hctsa achieves significant dimensionality reduction, making it suitable for large datasets and long time series.
- Speed: Once trained, hctsa can classify new time series very quickly, making it appropriate for real-time applications.

Despite its strengths, hctsa also face the following limitations:

- Computational Cost: The initial training of hctsa classifiers can be computationally expensive due to the large number of features calculated.
- Sensitivity to Data Imbalance: hctsa classifiers can be sensitive to class imbalances in the training data.
- Dependence on Feature Library: The success of hctsa relies heavily on the comprehensiveness of its feature library. While the current library is extensive, it may not cover all possible time-series features.
- Potential for Overfitting: Complex classifiers built using hctsa may be prone to overfitting, especially with limited training data. Cross-validation can be used to mitigate this, but is problematic for datasets with few training examples.

22 Canonical Time-series CHaracteristics (catch22)

The catch22 feature set was derived from the hctsa (highly comparative time-series analysis) toolbox, which contains over 7,000 features []. catch22 was designed to be a smaller and more efficient subset of features that still captures the diversity of analysis approaches contained in hctsa]. The 22 features were selected by evaluating their classification accuracy on 93 time-series classification datasets. The features were then clustered based on their performance, and the best-performing feature from each cluster was selected.

A description of each of the 22 features in the catch22 set, grouped by the type of time-series characteristic they measure, are given in details in https://time-series-features.gitbook.io/catch22 and presented more briefly in the following:

Distribution Shape

1. **DN_HistogramMode_5**: This feature calculates the mode of the distribution of values in the time series using a 5-bin histogram. The time series is first z-score normalized, meaning that the values are transformed to have a mean of 0 and a standard deviation of 1.
2. **DN_HistogramMode_10**: This is the same as the previous feature but uses a 10-bin histogram.

Extreme Event Timing

3. **DN_OutlierInclude_p_001_mdrmd**: This feature measures the time intervals between successive extreme events above the mean of the time series. An extreme event is defined as a value that is greater than or equal to a certain number of standard deviations above the mean. In this case, the threshold is set to 0.01 standard deviations above the mean. The feature then calculates the median of the distribution of these time intervals.
4. **DN_OutlierInclude_n_001_mdrmd**: This is the same as the previous feature, but it measures the time intervals between successive extreme events below the mean.

Linear Autocorrelation Structure

5. **first1e_acf_tau**: This feature finds the first 1/e crossing of the autocorrelation function. The autocorrelation function measures the correlation between a time series and a lagged version of itself. The 1/e crossing is the lag at which the autocorrelation function first falls to 1/e of its maximum value.

6. **firstMin_acf**: This feature finds the first minimum of the autocorrelation function.
7. **SP_Summaries_welch_rect_area_5_1**: This feature calculates the total power in the lowest fifth of frequencies in the Fourier power spectrum of the time series. The Fourier power spectrum decomposes the time series into a sum of sine waves of different frequencies. This feature measures the amount of power in the low-frequency components of the signal.
8. **SP_Summaries_welch_rect_centroid**: This feature calculates the centroid of the Fourier power spectrum. The centroid is a measure of the "center of mass" of the power spectrum. It is calculated as the weighted average of the frequencies in the power spectrum, where the weights are the power at each frequency.

Simple Forecasting

9. **FC_LocalSimple_mean3_stderr**: This feature calculates the mean error from a rolling 3-sample mean forecasting of the time series. The 3-sample mean forecasting is a simple forecasting method that predicts the next value in the time series by averaging the previous three values. This feature calculates the mean of the absolute differences between the predicted values and the actual values.

Incremental Differences

10. **FC_LocalSimple_mean1_tauresrat**: This feature calculates the change in correlation length after iteratively differencing the time series. Differencing a time series means subtracting each value from the previous value. The correlation length of a time series is a measure of how long it takes for the correlation between two points in the time series to decay to zero. This feature calculates the ratio of the correlation length of the differenced time series to the correlation length of the original time series.
11. **MD_hrv_classic_pnn40**: This feature calculates the proportion of successive differences that exceed 0.04 standard deviations of the time series. This is a measure of heart rate variability (HRV) that has been used to study the autonomic nervous system.

Symbolic

12. **SB_BinaryStats_mean_longstretch1**: This feature computes the longest sequence of successive values in the time series that are greater than the mean.
13. **SB_BinaryStats_diff_longstretch0**: This feature calculates the longest period of successive incremental decreases in the time series.
14. **SB_MotifThree_quantile_hh**: This feature first symbolizes the time series using a 3-symbol alphabet [12]. The alphabet is defined by dividing the range of values in the time series into three quantiles. The feature then calculates the Shannon entropy of the distribution of two successive symbols in the symbolized time series.

Nonlinear Autocorrelation

15. **CO_HistogramAMI_even_2_5**: This feature calculates the automutual information (AMI) of the time series. The AMI is a measure of how much information is shared between a time series and a lagged version of itself. It is calculated by first symbolizing the time series, meaning that each value is replaced with a symbol from a finite alphabet. In this case, the time series is symbolized using a 2-symbol

alphabet, and the lag is set to 5. The feature then calculates the Shannon entropy of the distribution of symbol pairs.

16. **CO_trev_1_num**: This feature calculates the time-reversibility statistic, which is a measure of how symmetric the time series is in time. A time series is time-reversible if it looks the same whether it is played forward or backward in time. This feature calculates the mean of the cubed differences between consecutive values in the time series.
17. **IN_AutoMutualInfoStats_40_gaussian_fmmi**: This feature finds the first minimum of the automutual information function. The automutual information function is a plot of the AMI as a function of the lag. The first minimum of this function is the lag at which the AMI first reaches a minimum value.

Self-Affine Scaling

18. **SC_FluctAnal_2_dfa_50_1_2_logi_prop_r1**: This feature calculates the proportion of slower timescale fluctuations in the time series that scale according to detrended fluctuation analysis (DFA). DFA is a method for quantifying the long-range correlations in a time series. The feature calculates the proportion of time scales for which the DFA fluctuation function scales with a power-law exponent greater than 1.
19. **SC_FluctAnal_2_rsrangefit_50_1_logi_prop_r1**: This feature calculates the proportion of slower timescale fluctuations in the time series that scale with linearly rescaled range fits. This is another method for quantifying the long-range correlations in a time series.

Other

20. **SB_TransitionMatrix_3ac_sumdiagcov**: This feature first symbolizes the time series using a 3-symbol alphabet. The alphabet is defined by calculating the autocorrelation function of the time series and dividing it into three quantiles. The feature then calculates the sum of the diagonal elements of the covariance matrix of the transition matrix between symbols.
21. **CO_Embed2_Dist_tau_d_expfit_meandiff**: This feature calculates an exponential fit to the distribution of successive distances between points in a 2-D embedding space of the time series. The embedding space is created by plotting each value in the time series against the previous value. The feature then calculates the mean difference between the fitted exponential distribution and the actual distribution of distances.
22. **PD_PeriodicityWang_th0_01**: This feature is a periodicity measure from Wang et al. (2007). It measures the strength of the periodicity in the time series.

The catch22 feature set can be computed on a time series of 10,000 samples in less than 0.5 seconds using the C implementation. This makes catch22 suitable for analyzing long time series that may be too computationally expensive to analyze using the full hctsa feature set. Additionally, feature calculation can be parallelized to further improve computational speed.

- **hctsa** (highly comparative time-series analysis) is a software tool that applies massive feature extraction to identify informative and interpretable quantitative phenotypes from time-series data [4, 5]. It uses over 7,700 features derived from the scientific time-series analysis literature, including features from the time domain, frequency domain, and time-frequency domain [4, 6].

LEFT TO DO:

1. To write:

- Find references for time-domain, frequency-domain, wavelets
- Review of applications with catch 22
- Description of DB that we will use (BitBrains, Time Series Classification Website)
- Catch22 application (use db to find window, justify selection)
- Apply anomaly detection method on catch22 features (RNN, AE)
- Put KAN in the game

2. To formalize:

- cross-references
- put in order references
- write in Latex

Literature review sources

Most of the references in the literature review were obtained from the following digital databases for scientific research:

1. Scopus (https://www.scopus.com/)
2. Google Scholar (https://scholar.google.com/)
3. IEEE Xplore (https://ieeexplore.ieee.org/)
4. Science Direct (https://www.sciencedirect.com/)
5. Springer Link (https://link.springer.com/)
6. IEEE Computer Society Digital library (https://www.computer.org/csdl/home)
7. PubMed (https://pubmed.ncbi.nlm.nih.gov/)
8. BMC (https://www.biomedcentral.com/)

References

1. Hadi Hojjati, Thi Kieu Khanh Ho, Narges Armanfard, Self-supervised anomaly detection in computer vision and beyond: A survey and outlook, Neural Networks, Volume 172, 2024, 106106, ISSN 0893-6080, https://doi.org/10.1016/j.neunet.2024.106106.
2. K. U. Duja, I. A. Khan and M. Alsuhaibani, "Video Surveillance Anomaly Detection: A Review on Deep Learning Benchmarks," in IEEE Access, vol. 12, pp. 164811-164842, 2024, https://doi.org/10.1109/ACCESS.2024.3491868.
3. Belay, M.A.; Blakseth, S.S.; Rasheed, A.; Salvo Rossi, P. Unsupervised Anomaly Detection for IoT-Based Multivariate Time Series: Existing Solutions, Performance Analysis and Future Directions. Sensors 2023, 23, 2844. https://doi.org/10.3390/s23052844
4. Cavallaro, C., Cutello, V., Pavone, M., & Zito, F. (2023). Discovering anomalies in big data: A review focused on the application of metaheuristics and machine learning techniques. Frontiers in Big Data, 6, 1179625. https://doi.org/10.3389/fdata.2023.1179625
5. Arnaldo Sgueglia, Andrea Di Sorbo, Corrado Aaron Visaggio, Gerardo Canfora, A systematic literature review of IoT time series anomaly detection solutions, Future Generation Computer Systems, Volume 134, 2022, Pages 170-186, ISSN 0167-739X, https://doi.org/10.1016/j.future.2022.04.005.
6. Basora, L.; Olive, X.; Dubot, T. Recent Advances in Anomaly Detection Methods Applied to Aviation. Aerospace 2019, 6, 117. https://doi.org/10.3390/aerospace6110117
7. A. A. Cook, G. Mısırlı and Z. Fan, "Anomaly Detection for IoT Time-Series Data: A Survey," in *IEEE Internet of Things Journal*, vol. 7, no. 7, pp. 6481-6494, July 2020, https://doi.org/10.1109/JIOT.2019.2958185.
8. Shaukat, K. et al. (2021). A Review of Time-Series Anomaly Detection Techniques: A Step to Future Perspectives. In: Arai, K. (eds) Advances in Information and Communication. FICC 2021. Advances in Intelligent Systems and Computing, vol 1363. Springer, Cham. https://doi.org/10.1007/978-3-030-73100-7_60

9. Ane Blázquez-García, Angel Conde, Usue Mori, and Jose A. Lozano. 2021. A Review on Outlier/Anomaly Detection in Time Series Data. ACM Comput. Surv. 54, 3, Article 56 (April 2022), 33 pages. https://doi.org/10.1145/3444690
10. Zahra Zamanzadeh Darban, Geoffrey I. Webb, Shirui Pan, Charu Aggarwal, and Mahsa Salehi. 2024. Deep Learning for Time Series Anomaly Detection: A Survey. ACM Comput. Surv. 57, 1, Article 15 (January 2025), 42 pages. https://doi.org/10.1145/3691338.
11. K. Choi, J. Yi, C. Park and S. Yoon, "Deep Learning for Anomaly Detection in Time-Series Data: Review, Analysis, and Guidelines," in IEEE Access, vol. 9, pp. 120043-120065, 2021, https://doi.org/10.1109/ACCESS.2021.3107975.
12. H. -S. Wu, "A survey of research on anomaly detection for time series," 2016 13th International Computer Conference on Wavelet Active Media Technology and Information Processing (ICCWAMTIP), Chengdu, China, 2016, pp. 426-431, https://doi.org/10.1109/ICCWAMTIP.2016.8079887.
13. Qinghua Liu, Paul Boniol, Themis Palpanas, John Paparrizos. Time-Series Anomaly Detection: Overview and New Trends. Proceedings of the VLDB Endowment (PVLDB), 2024, 17 (12), pp.4229 4232. https://doi.org/10.14778/3685800.3685842.
14. Sebastian Schmidl, Phillip Wenig, and Thorsten Papenbrock. 2022. Anomaly detection in time series: a comprehensive evaluation. Proc. VLDB Endow. 15, 9 (May 2022), 1779–1797. https://doi.org/10.14778/3538598.3538602.
15. Gen Li, Jason J. Jung, Deep learning for anomaly detection in multivariate time series: Approaches, applications, and challenges, Information Fusion, Volume 91, 2023, Pages 93-102, ISSN 1566-2535, https://doi.org/10.1016/j.inffus.2022.10.008.
16. P. Yan et al., "A Comprehensive Survey of Deep Transfer Learning for Anomaly Detection in Industrial Time Series: Methods, Applications, and Directions," in IEEE Access, vol. 12, pp. 3768-3789, 2024, https://doi.org/10.1109/ACCESS.2023.3349132.
17. Nesryne Mejri, Laura Lopez-Fuentes, Kankana Roy, Pavel Chernakov, Enjie Ghorbel, Djamila Aouada, Unsupervised anomaly detection in time-series: An extensive evaluation and analysis of state-of-the-art methods, Expert Systems with Applications, Volume 256, 2024, 124922, ISSN 0957-4174, https://doi.org/10.1016/j.eswa.2024.124922.
18. Braei, M., & Wagner, S. (2020). Anomaly Detection in Univariate Time-series: A Survey on the State-of-the-Art. ArXiv. https://arxiv.org/abs/2004.00433.
19. U. A. Usmani, I. A. Aziz, J. Jaafar and J. Watada, "Deep Learning for Anomaly Detection in Time-Series Data: An Analysis of Techniques, Review of Applications, and Guidelines for Future Research," in IEEE Access, https://doi.org/10.1109/ACCESS.2024.3495819.
20. John Paparrizos, Paul Boniol, Themis Palpanas, Ruey S. Tsay, Aaron Elmore, and Michael J. Franklin. 2022. Volume under the surface: a new accuracy evaluation measure for time-series anomaly detection. Proc. VLDB Endow. 15, 11 (July 2022), 2774–2787. https://doi.org/10.14778/3551793.3551830.
21. Guansong Pang, Chunhua Shen, Longbing Cao, and Anton Van Den Hengel. 2021. Deep Learning for Anomaly Detection: A Review. ACM Comput. Surv. 54, 2, Article 38 (March 2022), 38 pages. https://doi.org/10.1145/3439950.
22. Galvão, Y.M., Castro, L., Ferreira, J. et al. Anomaly Detection in Smart Houses for Healthcare. SN COMPUT. SCI. 5, 136 (2024). https://doi.org/10.1007/s42979-023-02480-y.

23. Sakib, M., Mustajab, S. & Alam, M. Ensemble deep learning techniques for time series analysis: a comprehensive review, applications, open issues, challenges, and future directions. Cluster Comput 28, 73 (2025). https://doi.org/10.1007/s10586-024-04684-0
24. Hilal, W., Gadsden, S. A., & Yawney, J. (2022). Financial Fraud: A Review of Anomaly Detection Techniques and Recent Advances. Expert Systems With Applications, 193, 116429. https://doi.org/10.1016/j.eswa.2021.116429.
25. A. Kesharwani and P. Shukla, "A Review of Anomaly Detection using Machine Learning Techniques," 2024 1st International Conference on Advanced Computing and Emerging Technologies (ACET), Ghaziabad, India, 2024, pp. 1-6, https://doi.org/10.1109/ACET61898.2024.10730574.
26. X. Yang, X. Qi and X. Zhou, "Deep Learning Technologies for Time Series Anomaly Detection in Healthcare: A Review," in IEEE Access, vol. 11, pp. 117788-117799, 2023, https://doi.org/10.1109/ACCESS.2023.3325896.
27. M. Jin et al., "A Survey on Graph Neural Networks for Time Series: Forecasting, Classification, Imputation, and Anomaly Detection," in IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 46, no. 12, pp. 10466-10485, Dec. 2024, https://doi.org/10.1109/TPAMI.2024.3443141.
28. M. Khadiche, B. Boucheham and S. Bougueroua, "Techniques and Methods for Anomalies Detection in ECG as a Support for Medical Decision in Healthcare: A Review," 2023 International Conference on Decision Aid Sciences and Applications (DASA), Annaba, Algeria, 2023, pp. 187-191, https://doi.org/10.1109/DASA59624.2023.10286626.
29. Díaz-Cao, J.M., Liu, X., Kim, J. et al. Evaluation of the application of sequence data to the identification of outbreaks of disease using anomaly detection methods. Vet Res 54, 75 (2023). https://doi.org/10.1186/s13567-023-01197-3
30. A. Garg, W. Zhang, J. Samaran, R. Savitha and C. -S. Foo, "An Evaluation of Anomaly Detection and Diagnosis in Multivariate Time Series," in IEEE Transactions on Neural Networks and Learning Systems, vol. 33, no. 6, pp. 2508-2517, June 2022, https://doi.org/10.1109/TNNLS.2021.3105827.
31. Mayu Sakurada and Takehisa Yairi. 2014. Anomaly Detection Using Autoencoders with Nonlinear Dimensionality Reduction. In Proceedings of the MLSDA 2014 2nd Workshop on Machine Learning for Sensory Data Analysis (MLSDA'14). Association for Computing Machinery, New York, NY, USA, 4–11. https://doi.org/10.1145/2689746.2689747.
32. Chong Zhou and Randy C. Paffenroth. 2017. Anomaly Detection with Robust Deep Autoencoders. In Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '17). Association for Computing Machinery, New York, NY, USA, 665–674. https://doi.org/10.1145/3097983.3098052.
33. Julien Audibert, Pietro Michiardi, Frédéric Guyard, Sébastien Marti, and Maria A. Zuluaga. 2020. USAD: UnSupervised Anomaly Detection on Multivariate Time Series. In Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD '20). Association for Computing Machinery, New York, NY, USA, 3395–3404. https://doi.org/10.1145/3394486.3403392.
34. Kieu, Tung, B. Yang, Chenjuan Guo and Christian S. Jensen. “Outlier Detection for Time Series with Recurrent Autoencoder Ensembles.” International Joint Conference on Artificial Intelligence (2019).
35. An, J., & Cho, S. (2015). Variational Autoencoder based Anomaly Detection using Reconstruction Probability.

36. T. Kieu et al., "Anomaly Detection in Time Series with Robust Variational Quasi-Recurrent Autoencoders," 2022 IEEE 38th International Conference on Data Engineering (ICDE), Kuala Lumpur, Malaysia, 2022, pp. 1342-1354, https://doi.org/10.1109/ICDE53745.2022.00105.
37. Chen, N., Tu, H., Duan, X. et al. Semisupervised anomaly detection of multivariate time series based on a variational autoencoder. Appl Intell 53, 6074–6098 (2023). https://doi.org/10.1007/s10489-022-03829-1.
38. Ya Su, Youjian Zhao, Chenhao Niu, Rong Liu, Wei Sun, and Dan Pei. 2019. Robust Anomaly Detection for Multivariate Time Series through Stochastic Recurrent Neural Network. In Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD '19). Association for Computing Machinery, New York, NY, USA, 2828–2837. https://doi.org/10.1145/3292500.3330672.
39. S. Lin, R. Clark, R. Birke, S. Schönborn, N. Trigoni and S. Roberts, "Anomaly Detection for Time Series Using VAE-LSTM Hybrid Model," ICASSP 2020 - 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Barcelona, Spain, 2020, pp. 4322-4326, https://doi.org/10.1109/ICASSP40776.2020.9053558.
40. Benjamin Lindemann, Benjamin Maschler, Nada Sahlab, Michael Weyrich, A survey on anomaly detection for technical systems using LSTM networks, Computers in Industry, Volume 131, 2021, 103498, ISSN 0166-3615, https://doi.org/10.1016/j.compind.2021.103498.
41. Shen, Lifeng et al. “Timeseries Anomaly Detection using Temporal Hierarchical One-Class Network.” Neural Information Processing Systems (2020).
42. Pengfei Liu, Xiaoming Sun, Yang Han, Zhishuai He, Weifeng Zhang, Chenxu Wu, Arrhythmia classification of LSTM autoencoder based on time series anomaly detection, Biomedical Signal Processing and Control, Volume 71, Part B, 2022, 103228, ISSN 1746-8094, https://doi.org/10.1016/j.bspc.2021.103228.
43. Georgios Petmezas, Kostas Haris, Leandros Stefanopoulos, Vassilis Kilintzis, Andreas Tzavelis, John A Rogers, Aggelos K Katsaggelos, Nicos Maglaveras, Automated Atrial Fibrillation Detection using a Hybrid CNN-LSTM Network on Imbalanced ECG Datasets, Biomedical Signal Processing and Control, Volume 63, 2021, 102194, ISSN 1746-8094, https://doi.org/10.1016/j.bspc.2020.102194.
44. D. -H. Shin, R. C. Park and K. Chung, "Decision Boundary-Based Anomaly Detection Model Using Improved AnoGAN From ECG Data," in IEEE Access, vol. 8, pp. 108664-108674, 2020, https://doi.org/10.1109/ACCESS.2020.3000638.
45. Alamr, A.; Artoli, A. Unsupervised Transformer-Based Anomaly Detection in ECG Signals. Algorithms 2023, 16, 152. https://doi.org/10.3390/a16030152.
46. Shan, L., Li, Y., Jiang, H., Zhou, P., Niu, J., Liu, R., Wei, Y., Peng, J., Yu, H., Sha, X., & Chang, S. (2022). Abnormal ECG detection based on an adversarial autoencoder. Frontiers in Physiology, 13, 961724. https://doi.org/10.3389/fphys.2022.961724.
47. S. Liu et al., "Time Series Anomaly Detection With Adversarial Reconstruction Networks," in IEEE Transactions on Knowledge and Data Engineering, vol. 35, no. 4, pp. 4293-4306, 1 April 2023, https://doi.org/10.1109/TKDE.2021.3140058.
48. Tang, J., Li, J., Gao, Z., & Li, J. (2022). Rethinking Graph Neural Networks for Anomaly Detection. ArXiv. https://arxiv.org/abs/2205.15508

49. Y. Wu, H. -N. Dai and H. Tang, "Graph Neural Networks for Anomaly Detection in Industrial Internet of Things," in IEEE Internet of Things Journal, vol. 9, no. 12, pp. 9214-9231, 15 June15, 2022, https://doi.org/10.1109/JIOT.2021.3094295.
50. Lingqiang Xie, Dechang Pi, Xiangyan Zhang, Junfu Chen, Yi Luo, Wen Yu, Graph neural network approach for anomaly detection, Measurement, Volume 180, 2021, 109546, ISSN 0263-2241, https://doi.org/10.1016/j.measurement.2021.109546.
51. Deng, A., & Hooi, B. (2021). Graph Neural Network-Based Anomaly Detection in Multivariate Time Series. Proceedings of the AAAI Conference on Artificial Intelligence, 35(5), 4027-4035. https://doi.org/10.1609/aaai.v35i5.16523.
52. X. Ma et al., "A Comprehensive Survey on Graph Anomaly Detection With Deep Learning," in IEEE Transactions on Knowledge and Data Engineering, vol. 35, no. 12, pp. 12012-12038, 1 Dec. 2023, https://doi.org/10.1109/TKDE.2021.3118815.
53. Y. Liu, Z. Li, S. Pan, C. Gong, C. Zhou and G. Karypis, "Anomaly Detection on Attributed Networks via Contrastive Self-Supervised Learning," in IEEE Transactions on Neural Networks and Learning Systems, vol. 33, no. 6, pp. 2378-2392, June 2022, https://doi.org/10.1109/TNNLS.2021.3068344.
54. Wang, R., Liu, C., Mou, X., Gao, K., Guo, X., Liu, P., Wo, T., & Liu, X. (2022). Deep Contrastive One-Class Time Series Anomaly Detection. ArXiv. https://arxiv.org/abs/2207.01472.
55. C. Huang, Z. Wu, J. Wen, Y. Xu, Q. Jiang and Y. Wang, "Abnormal Event Detection Using Deep Contrastive Learning for Intelligent Video Surveillance System," in IEEE Transactions on Industrial Informatics, vol. 18, no. 8, pp. 5171-5179, Aug. 2022, https://doi.org/10.1109/TII.2021.3122801.
56. Reiss, T., & Hoshen, Y. (2023). Mean-Shifted Contrastive Loss for Anomaly Detection. Proceedings of the AAAI Conference on Artificial Intelligence, 37(2), 2155-2162. https://doi.org/10.1609/aaai.v37i2.25309.
57. Liu, Z., Wang, Y., Vaidya, S., Ruehle, F., Halverson, J., Soljačić, M., Hou, T. Y., & Tegmark, M. (2024). KAN: Kolmogorov-Arnold Networks. ArXiv. https://arxiv.org/abs/2404.19756.
58. Bozorgasl, Z., & Chen, H. (2024). Wav-KAN: Wavelet Kolmogorov-Arnold Networks. ArXiv. https://arxiv.org/abs/2405.12832.
59. Xu, K., Chen, L., & Wang, S. (2024). Kolmogorov-Arnold Networks for Time Series: Bridging Predictive Power and Interpretability. ArXiv. https://arxiv.org/abs/2406.02496.
60. Genet, R., & Inzirillo, H. (2024). TKAN: Temporal Kolmogorov-Arnold Networks. ArXiv. https://arxiv.org/abs/2405.07344.
61. J., C., Blanco, L., Pereira, R., & Caus, M. (2024). Kolmogorov-Arnold Networks (KANs) for Time Series Analysis. ArXiv. https://arxiv.org/abs/2405.08790.
62. Samadi, M. E., Müller, Y., & Schuppert, A. (2024). Smooth Kolmogorov Arnold networks enabling structural knowledge representation. ArXiv. https://arxiv.org/abs/2405.11318.
63. Lubba, C.H., Sethi, S.S., Knaute, P. et al. catch22: CAnonical Time-series CHaracteristics. Data Min Knowl Disc 33, 1821–1852 (2019). https://doi.org/10.1007/s10618-019-00647-x.
64. Fulcher Ben D., Little Max A. and Jones Nick S. 2013Highly comparative time-series analysis: the empirical structure of time series and their methods J. R. Soc. Interface.1020130048 http://doi.org/10.1098/rsif.2013.0048.
65. Fulcher, Ben & Jones, Nick. (2014). Highly Comparative Feature-Based Time-Series Classification. IEEE Transactions on Knowledge and Data Engineering. 26. http://doi.org/10.1109/TKDE.2014.2316504.

66. Ben D. Fulcher, Nick S. Jones, hctsa: A Computational Framework for Automated Time-Series Phenotyping Using Massive Feature Extraction, Cell Systems, Volume 5, Issue 5, 2017, Pages 527-531.e3, ISSN 2405-4712, https://doi.org/10.1016/j.cels.2017.10.001.